

Sep 23, 2026 · @Simon Kodibona

Rubric map

Every section of the CivicPulse notebooks is tied to a line of the DIRISA SDC 2026 marking rubric. Two notebooks are referenced: **RD** = `CivicPulse_RealData_2021.ipynb` (executed on the uploaded files) and **FP** = `CivicPulse_Gauteng_Analysis.ipynb` (full pipeline, runs once ward-level Census and 2016 results are added). Numbers below come from the RD run.

Rubric criterion	Where it is met	Status
Defined problem statement, why it matters	FP §1; this doc §Problem statement	Done
Data cleaning and processing	RD §1, §3; FP §3–§6	Done
Exploratory data analysis and visualisation	RD §4, §6; FP §8	Done
Feature engineering	RD §3.1, §6; FP §5.2–§5.6, §6	Done
Model selection, training, testing, evaluation	RD §7; FP §10–§12	Done
Model addresses the problem statement	RD §5–§7, §9; FP §9, §14	Partial — youth and deprivation need ward-level data
Interpretation of model outputs	RD §9; FP §13, §15.2	Done
Honest data limitations and mitigation	RD §1, §8; FP §1.4, §15.3	Done
Deployment in usable form	FP §14.3, §15.1 (artefacts for Streamlit)	App still to build
Code documentation	Markdown cell above every code cell in both notebooks	Done

Rubric criterion	Where it is met	Status
Video and slides	Day 6 of the plan	To do
Bonus: innovative techniques	Tent-station proxy, voting-district crosswalk, within-ward design, one-SE rule	Done

Problem statement

Question: To what extent do multi-dimensional deprivation (Census 2022) and ward-level party dominance (IEC 2016/2021) explain differences in voter participation, especially youth participation, across Johannesburg and Tshwane wards ahead of the 4 November 2026 elections?

The spec asks for a question that is specific, justified, cross-referenced and honest about limits. Each is met as follows.

Spec criterion	How CivicPulse meets it
Specific, well-scoped question	Two metros, one unit (the ward, $n = 242$), two named hypotheses (H1 deprivation friction, H2 incumbency demobilisation) with stated tests.
Why it matters, and to whom	IEC and civil society: where to aim registration and turnout drives. Parties: whether safe seats demobilise their own voters. Journalists: evidence beyond the "youth apathy" narrative. Municipalities: whether service failures carry a civic cost.
Combines more than one dataset	IEC results (2016, 2021), IEC 2026 registration by age, Stats SA Census 2022.
Honest account of limitations	Ward-level youth turnout is not published; FP §1.4 redefines the targets openly (overall turnout T1; youth registration T2).

Why this framing scores well: it replaces a vague topic ("youth apathy") with falsifiable claims. A negative result is still a finding, and RD already produced one for H2.

Data sources and collection

CivicPulse uses spec datasets 1 and 2 plus Census 2022 as the supplementary source the spec invites. This meets "combine or cross-reference more than one dataset".

Source	Spec item	Level	Status	Used for
IEC 2021 LGE results, Gauteng	Dataset 2	Voting district × party × ballot, 195,424 rows	Received, clean	Turnout, winners, margins, ENP, station types
IEC 2016 LGE results, Gauteng	Dataset 2	Same format expected	Not yet received	Retention, turnout change, lagged features
IEC voter registration by age, 2026	Dataset 1 (dashboard)	Ward or municipality	Not yet extracted	Youth registration share and rate (T2)
IEC by-elections 2021–2026	Dataset 2	Ward	Optional	Mid-term turnout signal
Stats SA Census 2022: water, dwelling, refuse, age	Supplementary	Received at municipality level; ward level needed	Re-export required	Deprivation features (H1)

Collection justification. Downloadable IEC files are used wherever they exist, because they are complete and reproducible. The registration dashboard needs extraction per selection, as the spec notes; the notebook accepts a ward × age × gender CSV. Census files come from the Stats SA portal export, whose `geoLevelDesc` column records the geography level, and FP now refuses non-ward files with a clear message.

All supplementary data must be cited in the slides: Statistics South Africa, Census 2022; Electoral Commission of South Africa, results.elections.org.za.

Data cleaning and processing

The rubric names three things: missing values, inconsistent formats, and merging sources. Each cleaning step targets one of them, and each fixes a real problem found in the data or in the original loader.

Step	Rubric element	Problem it solves	Evidence
Data audit before any transform (RD §1)	Inconsistent formats	Detects geography level, empty columns and duplicates	Found all 8 census files at municipality level; Copy_of_AgeGroups_Johannesburg.csv identical by MD5; water counts = "undefined"
Metro filter by municipality code JHB/TSH, ward prefix 798/799	Inconsistent formats	Original filter GT481 GT484 selected Mogale City and Merafong	242 wards kept: 135 JHB, 107 TSH, matching official counts
Ward ID standardisation ("Ward 79800001" → 79800001)	Inconsistent formats	Mixed text and numeric IDs across sources	FP handles 8-digit IDs, short ward numbers and Stats SA labels
Keep Ward ballot only	Cleaning	PR and DC ballots are separate votes by the same people	45,086 of 195,424 rows retained for the metros
De-duplicate registered and spoilt per voting district	Cleaning	These repeat on every party row; summing would inflate them roughly 30×	One value per voting district
Exclude impossible voting districts	Missing/invalid values	Votes exceeding the roll	5 of 1,645 excluded, e.g. 605 valid votes vs 380 registered
Aggregate before merging	Merging sources	Long tables merged directly cause many-to-many row explosion	Every table reduced to one row per ward first
Left joins from the 2021 ward universe	Merging, missing values	Inner joins silently dropped wards	Missing pieces stay visible as NaN; FP reports coverage

Step	Rubric element	Problem it solves	Evidence
Imputation inside the model pipeline	Missing values	Imputing the whole table first leaks test information	Median imputation + missing-indicator, fitted per training fold
2016 → 2021 voting-district crosswalk (FP §5.5)	Merging across time	Wards redrawn: JHB 130 → 135, TSH 105 → 107	Coverage printed; naive ID match only as a flagged fallback

Why it matters for marking: each step is visible in a notebook cell with its before/after counts, so the cleaning is reproducible and auditable, not asserted.

Feature engineering

The rubric's own examples are turnout rates, registration gaps, fragmentation indices and geographic features. CivicPulse builds all four types, and each feature exists because a hypothesis needs it.

Turnout and fragmentation are defined as:

$$\text{turnout} = \frac{\text{valid} + \text{spoilt}}{\text{registered}}$$

$$\text{ENP} = \frac{1}{\sum_i s_i^2}$$

Feature	Rubric type	Built from	Why it is needed	Notebook
turnout (ward and voting district)	Turnout rate	IEC 2021	Primary target T1	RD §3
winner, winner_share, margin	Representation	IEC party shares	Dominance measure for H2	RD §3.1
enp (effective number of parties)	Fragmentation index	IEC party shares	Competitiveness; 1 = one-party	RD §3.1
share_ANC/DA/EFF/ActionSA	Representation	IEC	Controls for who dominates	RD §3.1
spoilt_rate	Participation quality	IEC	Protest or confusion signal	RD §3
tent, station_type, tent_share	Geographic proxy	Voting station names	Informal-settlement marker at sub-ward scale	RD §6

Feature	Rubric type	Built from	Why it is needed	Notebook
<code>log_registered</code>	Scale	IEC	Controls for VD size	RD §6
<code>retained</code> , <code>anc_retained</code>	Incumbency	IEC 2016 + 2021	H2: same party both terms	FP §6
<code>d_turnout</code>	Change	IEC 2016 + 2021	Disengagement signal	FP §6
<code>d_incumbent_share</code>	Change	IEC 2016 + 2021	Defection signal; pairs with <code>d_turnout</code>	FP §6
<code>pct_water_outside_dwelling</code> , <code>pct_informal_dwelling</code> , <code>pct_refuse_not_weekly</code>	Deprivation shares	Census 2022	H1 components	FP §5.2
<code>dep_index_z</code> , <code>dep_index_pca</code>	Composite index	Census components	One H1 effect size; PCA checks it	FP §6
<code>youth_share_adults</code>	Demographic	Census age bands	Youth weight per ward	FP §5.3
<code>pop_18_29_2026</code>	Registration gap denominator	Census aged 14–25 in 2022	Ages the cohort four years to election day	FP §5.3
<code>youth_reg_share_2026</code> , <code>youth_reg_rate_2026</code>	Registration gap	IEC registration + Census	Youth target T2	FP §5.6
<code>lag_*</code> features	Temporal	Previous election	Enables an honest 2026 forecast	FP §10.1

Design choices worth stating in the video. Only one water indicator enters the index, to avoid double-weighting water. Age bands are split proportionally (18–19 = 2/5 of the 15–19 band). The deprivation index is excluded from the predictive model because it duplicates its own components.

Exploratory data analysis and visualisation

The EDA's headline: in 2021 the turnout divide ran along party lines, not metro lines. Every plot answers one question that feeds a later decision.

Visual	Question it answers	Real finding (RD)	Decision it informs
Ward and VD turnout histograms by metro	How is turnout distributed?	Ward turnout 21.1%–65.7%, mean 42.8%, near-symmetric (skew 0.17)	Regression target is well-behaved
Metro comparison	Is metro a major driver?	Overall 42.0% JHB vs 44.6% TSH; ward medians 42.1% vs 41.1%, not significant	Metro kept as a control, not a focus
Boxplot by 2021 winner	Does turnout differ by party?	Median 39.5% in ANC wards vs 51.7% in DA wards	Winner party must be controlled in H2
Margin and ENP scatterplots	Does dominance track turnout?	Raw pattern: safer seats, higher turnout	Motivates the within-party test
Spearman heatmap	Which features co-move with turnout?	EFF share $\rho = -0.78$; DA share $\rho = +0.61$	Party shares enter the model
Lowest/highest 10 wards	Where are the extremes?	All 10 lowest are ANC-won; all 10 highest are DA-won Tshwane wards	Targets for the app's ward view
Margin vs turnout by party and metro (Implot)	Is the dominance effect uniform?	Slope only in DA wards	Interaction model M3
Station-type boxplot; within-ward tent scatter	Do informal areas vote less?	Tent VDs about 1 pp lower within wards, not significant	Proxy labelled exploratory
Predicted vs actual; permutation importance	Does the model fit, and on what?	Out-of-fold R^2 0.37	Model credibility

Why this satisfies the rubric: the plots are chosen to test assumptions and drive modelling choices, not to decorate. Each is followed by a markdown cell saying what to read from it.

Statistical analysis

Each test was chosen for a property of this data: small n, skewed features, correlated predictors, many simultaneous comparisons. The strongest result is that dominance goes with higher, not lower, turnout within DA wards, and has no effect in ANC wards.

Test	Why this test	Result (RD)
Shapiro-Wilk on ward turnout	Decide parametric vs rank methods	W = 0.980, p = 0.002: mild non-normality, so rank tests for group comparisons
Mann-Whitney U, JHB vs TSH	Two groups, no normality assumption	p = 0.23, rank-biserial 0.09: no metro difference
Kruskal-Wallis by winner group	More than two groups, rank-based	H = 106.9, p < 10 ⁻²³ : party is a major split
Spearman correlations + Benjamini-Hochberg FDR	Robust to skew; controls false discoveries across 9 tests	8 of 9 significant at q < 0.05
VIF	Detect collinear predictors	ANC and DA shares VIF ≈ 11: justifies Ridge over OLS
Standardised OLS with HC3 robust SEs (M1, M2, M4)	Comparable effect sizes; valid under unequal variance	Margin: +0.35 SD raw; +0.22 SD within same winner (p < 0.001)
Interaction model M3	Does the effect differ by party?	ANC wards +0.05 SD (p = 0.58); DA wards +0.41 SD (p < 0.001)
Within-ward fixed effects, ward-clustered SEs	Hold ward politics constant for the tent proxy	Tent -1.0 pp, 95% CI [-3.1, +1.1], p = 0.36
Cronbach's α, PCA (FP)	Validate the deprivation index	Runs once ward Census data arrives

Test	Why this test	Result (RD)
Bootstrap of the difference in effect sizes (FP)	H1 compares two coefficients directly	Needs ward-level income, if available
Breusch–Pagan (FP)	Check heteroscedasticity	Supports the HC3 choice

Caution to state: these are cross-sectional associations. Low turnout among one group can itself change the margin, so M2/M3 do not prove causation.

Model selection, training, testing and evaluation

Ridge regression was selected. It predicts voting-district turnout within 6.3 percentage points on wards it never saw, versus 8.6 points for predicting the average.

Regression or classification? Regression is primary: turnout is continuous and the hypotheses concern effect sizes. Classification (bottom-tercile "low-turnout" flag) is secondary, for targeting campaigns. The data settled which to deploy: ranking by Ridge predictions gives AUC 0.76, matching dedicated classifiers (0.75–0.77). One regression model serves both uses.

Model	Why it is a candidate	MAE (pp)	R ²
Baseline (mean)	Floor every model must beat	8.57	0.00
OLS	Transparent benchmark	6.33	0.365
Ridge (chosen)	Shrinkage for collinear party shares	6.33	0.365
Random Forest	Non-linearity, interactions	6.28	0.373
Gradient Boosting	Strong tabular learner	6.55	0.308
XGBoost	Named in the proposal	6.50	0.348

Values are held-out-ward means over 5 folds (RD §7.1). Classification AUC: logistic 0.746, random forest 0.769, Ridge-as-ranker 0.757.

Evaluation choice	Justification
GroupKFold by ward (5 folds)	All voting districts of a ward are held out together, so the model cannot learn a ward from its own neighbours
MAE in percentage points	Easiest error to explain to a non-technical panel
R ² and fold standard deviations	Shows explained variance and stability
One-standard-error rule	Picks the simplest model statistically tied with the best (Random Forest 6.28 ± 0.27 ; Ridge 6.32 falls inside)
Stability rule: exclude OLS if VIF > 10	OLS coefficients swing when features nearly sum to one; max VIF was 11.2
AUC, balanced accuracy (classification)	Ranking quality and performance on an imbalanced flag

FP extends this with repeated spatial-block CV, a random-vs-spatial CV comparison to measure optimism, and a lagged design (train 2016 → 2021, predict 2021 → 2026), which answers the rubric's note on historical data.

Interpretation against the problem statement

On current data, political geography explains about 37% of turnout variation; the deprivation question is still open until ward-level Census data is in. The rubric asks for outputs read back against the original question, so each hypothesis gets a verdict.

Hypothesis	Evidence so far	Verdict	What would change it
H2 — dominance demobilises	Within the same winning party, a 1 SD larger margin goes with +0.22 SD turnout; ANC wards show no margin effect ($p = 0.58$)	Not supported cross-sectionally	2016 file: test retention and defection vs disengagement directly

Hypothesis	Evidence so far	Verdict	What would change it
H1 — deprivation friction	Tent proxy: -1.0 pp within wards, not significant	Inconclusive	Ward-level Census 2022 water, dwelling, refuse shares
Youth gap	Metro level only: 18-34 is 46.1% of adults in JHB, 43.7% in TSH	Not yet testable at ward level	IEC 2026 registration by age per ward

Model outputs, in plain words. The strongest predictors are ANC share, DA share, ENP and EFF share. Turnout is lowest where ANC and EFF support is concentrated, and highest in DA strongholds in Tshwane. The model's error of 6.3 points is small relative to the 44-point spread between the lowest and highest wards.

The story for the presentation. The ANC-DA turnout gap is about 12 points. The next step asks how much of that gap is deprivation rather than party — H1 in its sharpest form.

Data limitations and mitigation

The rubric rewards naming limitations and showing how each was handled. CivicPulse lists them in the notebooks themselves (RD §1 and §8, FP §1.4 and §15.3).

Limitation	Effect on the analysis	Mitigation
Ward-level youth turnout is not published	Cannot model youth turnout directly	Two explicit targets: overall turnout (T1) and youth registration share/rate (T2)
Census files exported at municipality level	Only 2 data points; H1 untestable	Audit caught it; re-export at ward level; FP rejects non-ward files
Registration data is a dashboard snapshot	Extraction effort; no time series	Pair with 2016/2021 results and by-elections for temporal signal
Only two elections	"Prolonged tenure" is coarse	Define as held in both 2016 and 2021; add 2011 via the crosswalk if time allows

Limitation	Effect on the analysis	Mitigation
Ward boundaries changed 2016 → 2021	Same ID is not the same area	Voting-district crosswalk with printed coverage
Census 2022 sits between elections	Timing mismatch	Treat deprivation as slow-moving structure
Ward-level income may not exist for Census 2022	H1 income comparison may be impossible	Check availability; otherwise use a cited proxy and say so
Small sample (242 wards)	Wide intervals, overfitting risk	Regularised models, confidence intervals, one-SE rule, VD-level models (n = 1,640)
Spatial dependence	Optimistic scores under random CV	Group CV by ward/spatial block; FP measures the optimism gap
Ecological inference	Ward patterns ≠ individual behaviour	Stated in every conclusion
Tent proxy is indirect	Captures only extreme informality	Labelled exploratory; to be validated against Census informal-dwelling shares
5 VDs with votes above the roll	Impossible turnout values	Excluded and listed in the notebook

Deployment, code documentation and presentation

Deployment. The rubric wants a usable form, not a static script. FP already exports everything the app needs: `turnout_model.joblib`, `model_metadata.json`, `master_ward_table.csv` and `forecast_2026.csv`. The planned Streamlit app has three views, each tied to an audience from the problem statement.

App view	Audience	Backed by
Ward explorer: turnout, winner, margin, ENP, deprivation	Journalists, voters	Master ward table

App view	Audience	Backed by
Priority list: low forecast turnout × low youth registration	IEC, civil society	Forecast + registration (FP §14.2)
Policy simulator: change deprivation inputs, see predicted turnout shift	Policymakers	<code>simulate_policy</code> (FP §14.3), labelled as association, not cause

Code documentation. Every code cell has a markdown cell above it stating what it does and why. Configuration sits in one cell. Both notebooks run top to bottom without errors, and fixed random seeds make results reproducible. For the submission, add a short package list (pandas, numpy, scipy, statsmodels, scikit-learn, xgboost, matplotlib, seaborn) with versions.

Presentation. The 15-minute video must include every team member. A split that follows this document keeps each speaker on one rubric area:

Speaker	Content	Minutes
1	Problem, why it matters, data sources	3
2	Cleaning and feature engineering	3
3	EDA and statistical tests	3
4	Model, evaluation, interpretation	3
5	App demo, limitations, recommendations	3

Slides must cite every source (IEC, Stats SA) and packages used.

Bonus: innovative techniques

The rubric's bonus examples are creative geographic features, novel cross-referencing, and unconventional but justified modelling. CivicPulse has one of each, plus a methodological safeguard.

Technique	Bonus category	Why it is novel and justified
Tent voting stations as an informal-settlement marker	Creative feature engineering	Derived from station names in the IEC file itself; 155 of 1,640 VDs; gives sub-ward deprivation signal without external data
Within-ward fixed-effects comparison	Unconventional, justified modelling	Compares tent and building VDs in the same ward (99 wards), holding politics and councillor constant
Voting-district crosswalk for 2016 → 2021	Novel cross-referencing	Re-aggregates old results onto new ward boundaries instead of matching IDs that no longer mean the same area
Defection vs disengagement decomposition	Novel cross-referencing	Separates incumbent-share change from turnout change to test H2's mechanism, not just its outcome
Registration rate with an aged Census cohort	Novel cross-referencing	Uses Census people aged 14–25 in 2022 as the 18–29 electorate on 4 November 2026
One-SE rule + VIF stability rule, fixed before results	Justified modelling	Model choice is decided by pre-stated criteria, not by preference
Regression-as-ranker test	Justified modelling	Answers regression vs classification with data (AUC 0.757 vs 0.746–0.769)

Gaps to close and references

Submission is due on NextCloud by 08:00 on **Sep 28, 2026**. These items move rubric lines from Partial to Done, in priority order.

- ☐ Re-export the four Census 2022 indicators at Ward level for both metros (unlocks H1)

- ☐ Download the 2016 Gauteng LGE results in the same IEC format (unlocks H2 retention and the 2026 forecast)
- ☐ Extract 2026 registration by age per ward from the IEC dashboard (unlocks the youth target)
- ☐ Re-run FP on the complete data and paste the final numbers into this document and the slides
- ☐ Build the Streamlit app from the exported artefacts
- ☐ Add a package-and-version list to the notebook
- ☐ Record the 15-minute video with every member presenting; share the NextCloud link with bsebusho@csir.co.za

Open question: the speaker split above assumes five members; adjust to your team size.

References

- DIRISA SDC 2026, Teams Qualification Technical Document Specification (marking rubric).
- Electoral Commission of South Africa, 2021 Local Government Elections results, results.elections.org.za.
- Electoral Commission of South Africa, Voter Registration Statistics, elections.org.za.
- Statistics South Africa, Census 2022 (municipality-level exports received; ward-level to follow).
- Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning* (one-standard-error rule) — cited from memory; verify before use in slides.
- Python packages: pandas, numpy, scipy, statsmodels, scikit-learn, xgboost, matplotlib, seaborn.